

How to Explore Collatz-3

1. Introduction

Collatz-3 is a new, playful variant of the classic Collatz map, redesigned for the world of modulo 3.

This document introduces the rules, behavior, and structural features of the Collatz-3 system, along with its “3-Resistance” landscape for integers from 1 to 100.

Although the rule is simple, the dynamics are surprisingly rich. Some numbers fall quickly toward 1, while others wander through long, chaotic trajectories before finally reaching the end.

This tendency to “resist” the modulo-3 world is what we call **3-Resistance**, and the goal of the game is simple:

Find the number that resists 3 the most — the Boss.

2. Definition of the Collatz-3 Map

For any integer n , the Collatz-3 transformation is defined as:

$$n \equiv 0 \pmod{3} \rightarrow n / 3$$

$$n \equiv 1 \pmod{3} \rightarrow (4n - 1) / 3$$

$$n \equiv 2 \pmod{3} \rightarrow (4n + 1) / 3$$

The process continues until the value reaches **1**, at which point the sequence terminates.

This map alternates between:

- **division by 3** (rapid descent), and
- **$4n \pm 1$ transformations** (growth and oscillation),

creating a dance of rises and falls reminiscent of the classical Collatz map, but with a distinctly 3-adic flavor.

3. How to Play

1. Choose any positive integer n .
2. Apply the Collatz-3 rule repeatedly.
3. Count the number of steps until the sequence reaches 1.
4. The more steps it takes, the stronger its 3-Resistance.

Objective:

Identify the integer with the highest 3-Resistance within a given range.

4. Example: The Legendary Boss "28"

Within the range 1–100, the number **28** exhibits the strongest 3-Resistance.

Its trajectory is long and intricate:

28 → 37 → 49 → 65 → 87 → 29 → 39 → 13 → 17 → 23 → 31 → 41 → 55 → 73 → 97
→ 129 → 43 → 57 → 19 → 25 → 33 → 11 → 15 → 5 → 7 → 9 → 3 → 1

Total steps: 27 This is the longest stopping time in the 1–100 range, making **28 the Boss**.

5. The 3-Resistance Landscape (1–100)

When the stopping time is visualized as height, the integers from 1 to 100 form a distinctive terrain:

Valleys (Low Resistance)

Multiples of 3 fall quickly due to the direct division by 3.

Ridges (High Resistance)

Numbers that repeatedly enter the $4n \pm 1$ cycle climb upward before descending.

The Isolated Peak

28 stands alone as the highest summit — a solitary mountain in the landscape.

This terrain reflects a unique structure different from the classical Collatz map, shaped by the arithmetic of modulo 3.

7. Conclusion

Collatz-3 is a simple yet remarkably expressive integer map. Its behavior reveals:

- rapid descent for multiples of 3,
- chaotic growth cycles for $4n \pm 1$ sequences,
- a landscape of valleys, ridges, and isolated peaks,
- and a clear Boss — **28**, the champion of 3-Resistance.

This system stands as an example of **playable mathematics**, blending curiosity, structure, and experimentation.

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